

*Programme Code(s): TU856, TU857, TU858*

*Shared With: NA*

*Module Code(s): CMPU3010*

*CRN: 28020, 31857, 34790*

**TECHNOLOGICAL UNIVERSITY DUBLIN**  
CITY CAMPUS, GRANGEGORMAN

---

TU856 – B.Sc.(Hons) Computer Science  
TU857 – B.Sc.(Hons) Computer Science (Infrastructure)  
TU858 – B.Sc.(Hons) Computer Science (International)

*Year 3*

---

*SEMESTER 1 2025/26*

---

***CMPU3010 Databases 2***

**Internal Examiner(s):**

Mr. Damian Gordon, Mr. Ciaran Kelly, Dr. Bianca Schoen-Phelan  
Dr. Brian Keegan

**External Examiner(s):**

Dr. Colm O’Riordan  
Ms. Caroline McEnroy

**Instructions To Candidates: Please answer all questions.**

**Exam Duration: 2 hours.**

**Special Instructions /Handouts/ Materials Required: None.**

## 1. Entities and Attributes

The Health Service Executive has hired you to develop a new Medical Practice System to collect, store, and manage details about medical practices, doctors and patients. You have identified five key entities in the system: Practice, Doctor, Patient, Prescription, and Appointment. They interact in the following ways:

- Each Practice employs many Doctors, but each Doctor can only work in one Practice.
- A Doctor can have many Appointments, but each Appointment is handled by only one Doctor.
- Each Appointment results in exactly one Prescription, and each Prescription belongs to only one Appointment.
- A Patient can have zero or many Appointments, but each Appointment must be associated with exactly one Patient.
- This structure allows doctors to manage multiple appointments, ensures every appointment generates only one prescription, and accounts for patients who may not have an appointment.

(a) Draw a *simple Entity Relationship Diagram* which highlights the relationships between each of the entities.

**(8 marks)**

(b) For each entity in part (a), add *attributes*, as well as *primary keys* and *foreign keys*, where appropriate.

**(7 marks)**

(c) Write suitable SQL code to create tables to represent the entities and attributes that you have described in part (b), ensuring that all the necessary primary keys are defined.

**(8 marks)**

(d) Using the JOIN command on the tables outlined in part (c), list the following:

- All Appointments in a given Practice
- All Prescriptions written by a given Doctor

**(4 marks)**

**Question 1) continues on the next page!**

1. (e) Explain what is meant by the terms ACID and BASE in the context of database transactions.

**(3 marks)**

- (f) The Health Service Executive's new Medical Practice System has introduced a feature to store patient medical information (such as allergies, current medications, and emergency contact details) as a JSON object inside the PATIENT table. Suggest the structure (and an example) of the JSON code.

**(3 marks)**

**Question 2) is on the next page!**

## 2. Data from around the organisation

- (a) A university keeps student and programme data in a relational database and has access to module review data from a social media system. The university wishes to use both sources of data to analyse module ratings at the end of each semester.

- i. Describe the differences between structured and unstructured data using examples from the scenario.

Explain the challenges of combining these data types in a single analysis pipeline.

**(5 marks)**

- ii. Demonstrate with examples how you would clean, transform, and store both types of data using appropriate tools (e.g., SQL for structured, Python for unstructured).

Include sample queries or code snippets that show how you would extract meaningful insights from each data type.

**(7 marks)**

- iii. The structured data set has the following format

- Student: (StudentNo, name, email, address1, address2)
- Modules: (ModuleCode, Module Name, ECTS, semester)
- Programme: (ProgrammeCode, ProgrammeName, startdate, enddate)
- Register: (StudentNo, Program Code, ModuleCode, Semester, Result (0-100%))

Propose a JSON data model for this data suitable for storing in MongoDB. Your model should consider embedding some repeating groups.

**(11 marks)**

- iv. Using MongoDB and JSON, design an input validation system for your model.

**(4 marks)**

- v. Using your datamodel, write MongoDB queries for the following :-

- Find the list of modules the student is studying.
- Find students and their modules with a result < 40%
- Count the number of students registered on a programme

**(6 marks)**

### 3. Query Performance

- (a) Define query optimisation **and** name two techniques used by database systems to improve query performance.  
(5 marks)

- (b) Describe how the query optimiser chooses an execution plan **and** explain why different plans may be generated for the same SQL query.  
(6 marks)

- (c) You run the following query on the Employees table:

```
SELECT employee_id, name, department
FROM employees
WHERE department = 'Sales';
```

You collect this execution plan (PostgreSQL-style):

```
Seq Scan on employees
(cost=0.00..215.00 rows=800 width=48)
(actual time=0.020..4.500 rows=800 loops=1)
  Filter: (department = 'Sales')
  Rows Removed by Filter: 9200
Planning Time: 0.120 ms
Execution Time: 4.700 ms
```

Based on this plan, describe how PostgreSQL is executing the query **and** suggest one concrete optimisation to improve performance. Explain how your improvement would likely change the plan.  
(7 marks)

- (d) A DBA is asked whether denormalizing several lookup tables into a single wide table would improve query performance.

Evaluate this proposal, outlining three advantages and three disadvantages, and conclude with a justified recommendation.

(15 marks)